Section 4–1 Boolean Operations and Expressions:

1. Using Boolean notation, write an expression that is a 1 whenever one or more of its variables (*A*, *B*, *C*, and *D*) are 1s.

3. Write an expression that is a 1 when one or more of its variables (*A*, *B*, and *C*) are 0s.

5. Find the values of the variables that make each product term 1 and each sum term 0.

(b) $A\bar{B}C$

(f) $\bar{A} + B$

6. Find the value of X for all possible values of the variables.

(b) $X = (\overline{A + B})C$

Section 4–2 Laws and Rules of Boolean Algebra:

7. Identify the law of Boolean algebra upon which each of the following equalities is based:

(b) $AB\bar{C}D + \overline{ABC} = D\bar{C}BA + \overline{CBA}$

(c) $AB(CD + E\bar{F} + GH) = ABCD + ABE\bar{F} + ABGH$

8. Identify the Boolean rule(s) on which each of the following equalities is based:

(b) $A\bar{A}B + AB\bar{C} + AB\bar{B} = AB\bar{C}$

(e) $A\bar{B} + A\bar{B}C = A\bar{B}$

(f) $ABC + \overline{AB} + \overline{ABC}D = ABC + \overline{AB} + D$

Section 4–3 DeMorgan's Theorems:

9. Apply DeMorgan's theorems to each expression:

(g) $\overline{AB + CD}$

10. Apply DeMorgan's theorems to each expression:

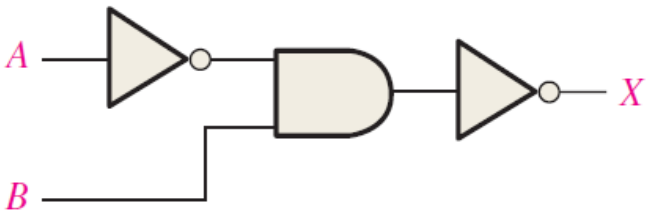
(c) $\overline{(A + \bar{B} + C + \bar{D})} + \overline{ABC\bar{D}}$

11. Apply DeMorgan's theorems to the following:

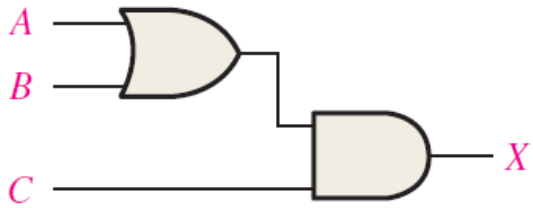
(b) $\overline{\left(A + \overline{B\bar{C}} + CD\right)} + \overline{\overline{BC}}$

Section 4–4 Boolean Analysis of Logic Circuits:

13. Write the Boolean expression for each of the logic circuits in Figure 4–57.



(c)



(d)

15. Draw the logic circuit represented by each expression:

(d) $A + B[C + D(B + \bar{C})]$

16. (b) Draw a logic circuit for the case where the output, HOLD, is HIGH only if the input, LOAD, is LOW and the input, READY, is HIGH.

18. Construct a truth table for each of the following Boolean expressions:

(e) $(A + B)(\bar{B} + C)$

Section 4–5 Logic Simplification Using Boolean Algebra:

19. Using Boolean algebra techniques, simplify the following expressions as much as possible:

(e) $A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C$

21. Using Boolean algebra, simplify the following expressions:

(c) $(B + BC)(B + \bar{B}C)(B + D)$

Section 4–6 Standard Forms of Boolean Expressions:

24. Convert the following expressions to sum-of-product (SOP) forms:

(c) $A + B[AC + (B + \bar{C})D]$

26. Convert each SOP expression in Problem 24 to standard SOP form.

(c) $A + B[AC + (B + \bar{C})D]$

30. Convert each standard SOP expression in Problem 26 to standard POS form.

(c) $A + B[AC + (B + \bar{C})D]$

Section 4–7 Boolean Expressions and Truth Tables:

35. Develop a truth table for each of the standard POS expressions:

(b) $(A+\bar{B})(A+\bar{B}+\bar{C})(B+C+\bar{D})(\bar{A}+B+\bar{C}+D)$

36. For each truth table in Table 4–15, derive a standard SOP and a standard POS expression.

(b) Table 4–15:

A	*B*	*C*	*X*
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Section 4–9 Karnaugh Map SOP Minimization:

40. Use a Karnaugh map to find the minimum SOP form for each expression:

(d) $\bar{A}\bar{B}\bar{C}+A\bar{B}\bar{C}+\bar{A}B\bar{C}+AB\bar{C}$

42. Expand each expression to a standard SOP form:

(c) $A\bar{B}\bar{C}D+AC\bar{D}+B\bar{C}D+\bar{A}BC\bar{D}$

43. Minimize each expression in Problem 42 with a Karnaugh map.

(c) $A\bar{B}\bar{C}D+AC\bar{D}+B\bar{C}D+\bar{A}BC\bar{D}$

46. Use the Karnaugh map method to implement the minimum SOP expression for the logic function specified in truth Table 4–17.

A	B	C	D	X
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0

1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

Ch. 4

$\boxed{1}$ Expression that is a 1 whenever 1^+ var's (A,B,C,D) are 1's.

$\Longrightarrow \quad A+B+C+D$

$\boxed{3}$ Expression that is a 1 when 1^+ var's (A,B,C) are 0's.

$\Longrightarrow \quad \bar{A}+\bar{B}+\bar{C}$

$\boxed{5}$ (b) $A\bar{B}C$ is product term

To make $A\bar{B}C=1 \Rightarrow A=1$ & $B=0$ & $C=1$

(f) $\bar{A}+B$ is sum term

To make $\bar{A}+B=0 \Rightarrow A=1$ & $B=0$

$\boxed{6}$ (b) $X = (\overline{A+B})\,C$

# of inputs = 3 $\Longrightarrow$ Truth Table has $2^3=8$ raws

A	B	C	A+B	$(\overline{A+B})$	$X=(\overline{A+B}).C$
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	0	0
1	1	1	1	0	0

$\boxed{7}$ (b) $AB\bar{C}D + \overline{ABC} = D\bar{C}BA + \overline{CBA}$

$\Longrightarrow$ Commutative Law

(c) $AB(CD + E\bar{F} + GH) = ABCD + ABE\bar{F} + ABGH$

$\Longrightarrow$ Distributive Law

$\boxed{8}$ (b) $A\bar{A}B + AB\bar{C} + AB\bar{B} = AB\bar{C}$

$\Longrightarrow$ Rule 8 : $X.\bar{X} = 0$

(e) $A\bar{B} + A\bar{B}C = A\bar{B}$

$\Longrightarrow$ Rule 10 : $X + XY = X$

(f) $ABC + \overline{AB} + \overline{ABC}D = ABC + \overline{AB} + D$

$\Longrightarrow$ Rule 11 : $X + \bar{X}Y = X + Y$

Ch4

9 (g) $\overline{AB+CD} = \overline{AB}\cdot\overline{CD} = (\bar{A}+\bar{B})(\bar{C}+\bar{D})$

---

10 (c) $\overline{(\overline{A+\bar{B}+C+\bar{D}})} + \overline{ABC\bar{D}}$

$= \bar{A}\,\bar{\bar{B}}\,\bar{C}\,\bar{\bar{D}} + \bar{A}+\bar{B}+\bar{C}+\bar{\bar{D}}$

$= \bar{A}B\bar{C}D + \bar{A}+\bar{B}+\bar{C}+D$

$= \bar{A}+\bar{B}+\bar{C}+D$

---

11 (b) $\overline{(A+\overline{B\bar{C}}+CD)} + \overline{\overline{BC}}$

$= \bar{A}\,(\overline{\overline{B\bar{C}}})\,(\overline{CD}) + BC$

$= \bar{A}\,B\bar{C}\,(\bar{C}+\bar{D}) + BC$

$= \bar{A}B\bar{C}\bar{C} + \bar{A}B\bar{C}\bar{D} + BC$

$= \underbrace{\bar{A}B\bar{C}} + \underbrace{\bar{A}B\bar{C}}\bar{D} + BC$

$= \bar{A}B\bar{C} + BC$

$= B(\bar{A}\bar{C}+C)$

$= B(\bar{A}+C)$

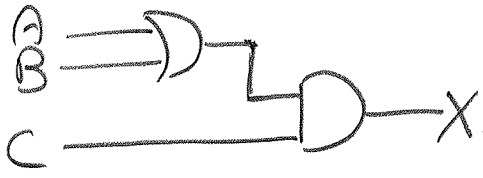
$= \bar{A}B + BC$

---

13 (c)

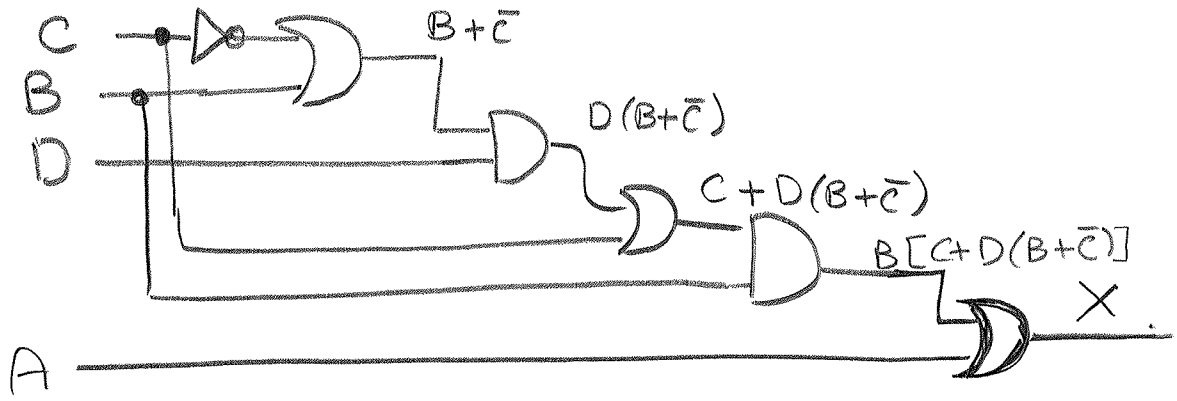
$X = (\overline{\bar{A}B})$

(d)

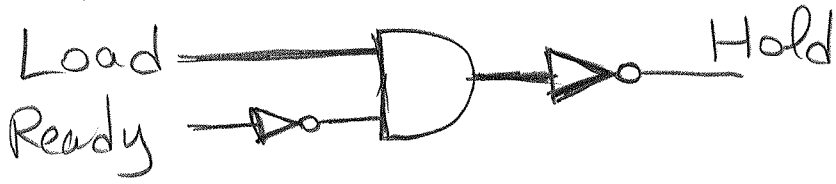


$X = (A+B).C$

15 (d) $X = A + B[C + D(B + \bar{C})]$



16 (b)



18 (e)

A	B	C	$A+B$	$\bar{B}+C$	$(A+B)(\bar{B}+C)$
0	0	0	0	1	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	1	0	0
1	1	1	1	1	1

19 (e) $A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C$

$= A\bar{B}C + \bar{A}C(B+\bar{B})$ [DL]

$= A\bar{B}C + \bar{A}C$ [R6, R4]

$= (A\bar{B} + \bar{A})C$ [DL]

$= (\bar{B} + \bar{A})C$ [R11]

$= \bar{B}C + \bar{A}C$ [DL]

21 (c) $(B+BC)(B+\bar{B}C)(B+D)$

$= B(B+C)(B+D)$ [R10 & R11]

$= B(B+CD)$ [R12]

$= BB + BCD$ [DL]

$= B + BCD$ [R7]

$= B$ [R10]

---

24 (c) $A + B[AC + (B+\bar{C})D]$ ← General Expression

$= A + ABC + B(B+\bar{C})D$

$= A + BBD + B\bar{C}D$

$= A + BD + B\bar{C}D$

$= A + BD$ ← non-standard SOP

---

26 (c) $A + BD$ ← non-standard SOP

$= A(B+\bar{B})(D+\bar{D}) + (A+\bar{A})BD$

$= ABD + AB\bar{D} + A\bar{B}D + A\bar{B}\bar{D} + \cancel{ABD} + \bar{A}BD$

$= ABD + AB\bar{D} + A\bar{B}D + A\bar{B}\bar{D} + \bar{A}BD$ ← standard SOP

---

30 (c)

standard-SOP

$X = ABD + AB\bar{D} + A\bar{B}D + A\bar{B}\bar{D} + \bar{A}BD$

A	B	D	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

standard-POS

$X = (A+B+\bar{D}) \cdot (A+B+\bar{D}) \cdot (A+\bar{B}+D)$

35 (b)

$$X = (A+\bar{B})(A+\bar{B}+\bar{C})(B+C+\bar{D})(\bar{A}+B+\bar{C}+D)$$

$$\Downarrow \qquad \Downarrow \qquad \Downarrow \qquad \Downarrow$$

0 1 x x 0 1 1 x x 0 0 1 1 0 1 0

A	B	C	D	
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

$(B+C+\bar{D})$

$(A+\bar{B})$

$(A+\bar{B}+\bar{C})$

$(\bar{A}+B+\bar{C}+D)$

---

36 (b) Standard POS

$$X = (A+B+C)\cdot(A+B+\bar{C})(A+\bar{B}+C)(A+\bar{B}+\bar{C})(\bar{A}+B+C)$$

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Standard SOP

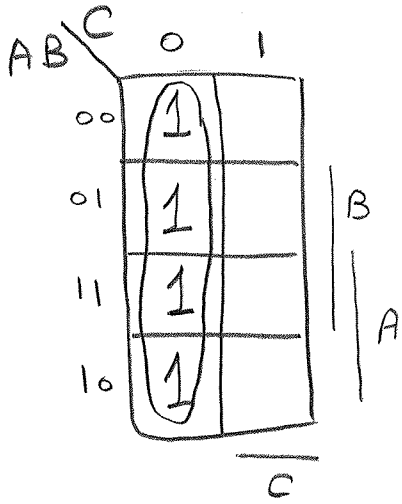
$$X = A\bar{B}C + AB\bar{C} + ABC$$

Ch4

40 (d) $X = \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C}$

000 100 010 110

$X = \bar{C}$



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42 (c) $X = A\bar{B}\bar{C}D + AC\bar{D} + B\bar{C}D + \bar{A}BC\bar{D}$

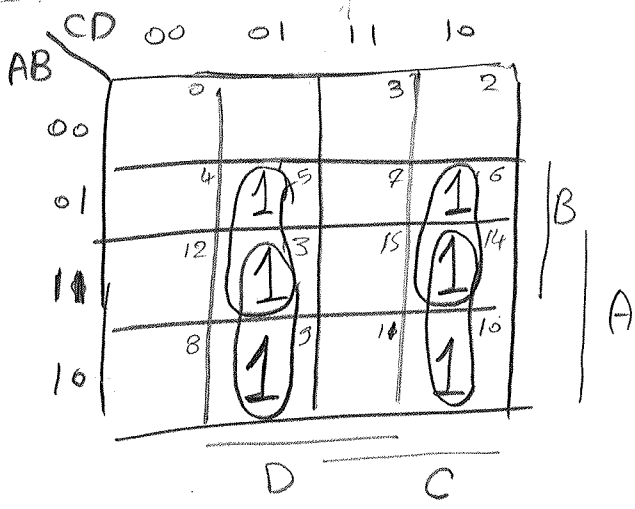
$= A\bar{B}\bar{C}D + A(B+\bar{B})C\bar{D} + (A+\bar{A})B\bar{C}D + \bar{A}BC\bar{D}$

$= A\bar{B}\bar{C}D + ABC\bar{D} + A\bar{B}C\bar{D} + ABC\bar{D} + \bar{A}B\bar{C}D + \bar{A}BC\bar{D}$

9 14 10 13 5 6

43 (c)

$X = B\bar{C}D + A\bar{C}D + BC\bar{D} + AC\bar{D}$



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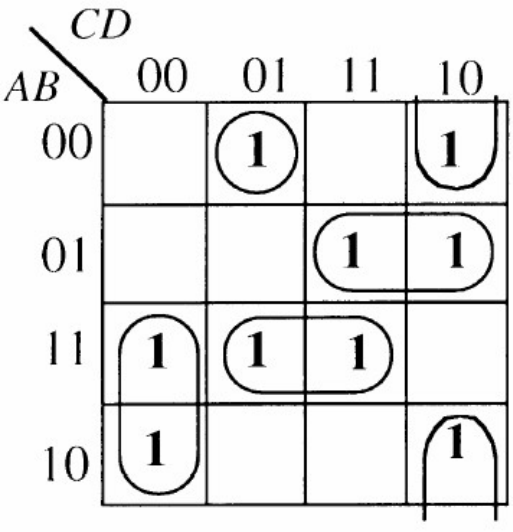
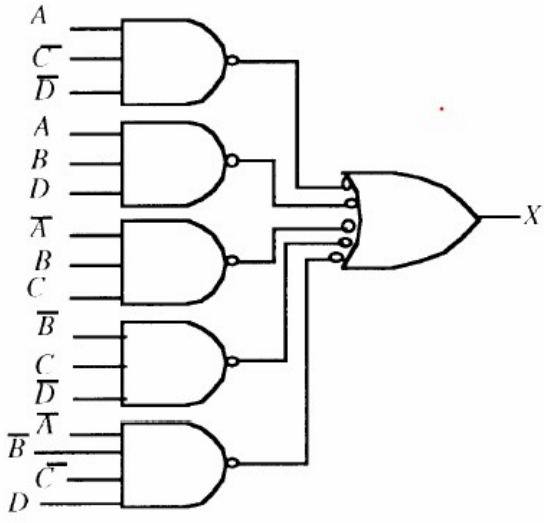
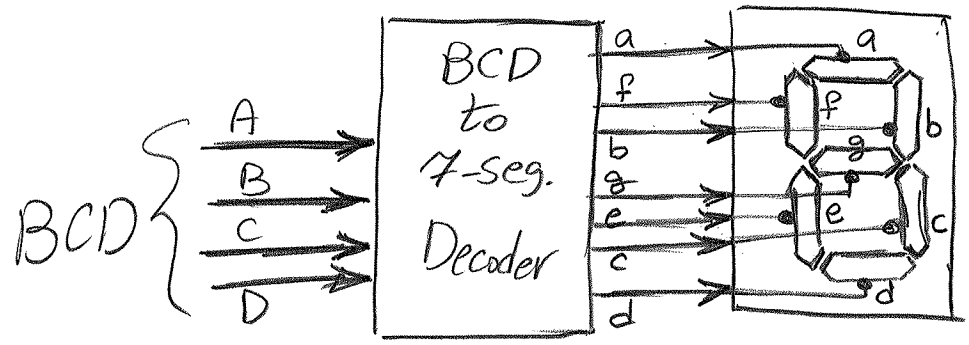


$X = A\bar{C}\bar{D} + ABD + \bar{A}BC + \bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$



Design Problem:

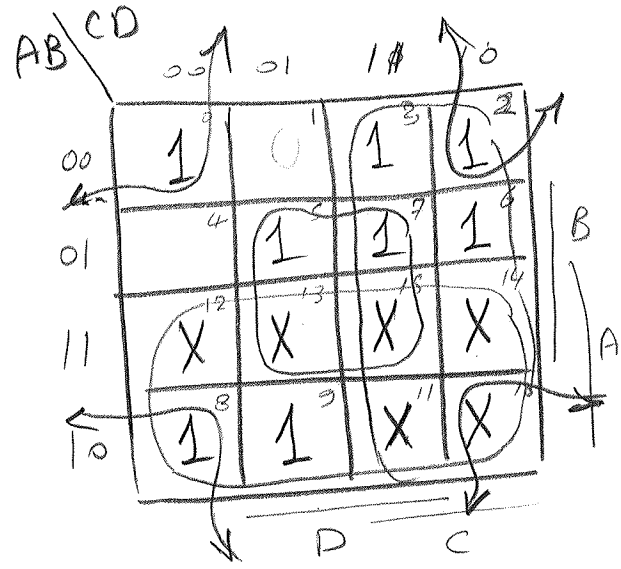
Design the logic circuit that controls the segment a



7-segment Display

(assuming common-cathod display $\Longrightarrow$ 1 turns segment on
0 turns segment off)

A	B	C	D	a
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	X
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X



$$a = A + C + BD + \bar{B}\bar{D}$$